a Leakage-aware training and evaluation

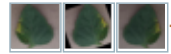


PlantVillage
3,500 images
7 classes
RGB 128 x 128

Leaf-ID union
1,799 groups

Train 2,448
Val 512
Test 540

Augmented run only



Flip Rotate Color
Base: no aug.

No random aug.

ImageNet normalize

ResNet18



ImageNet init.
Full fine-tuning

GAP
FC 7

7 logits

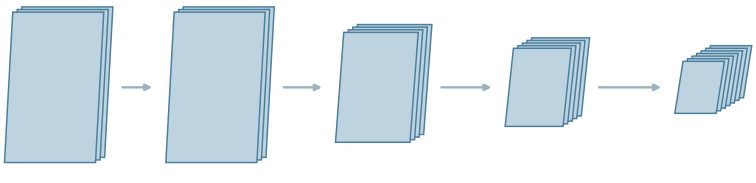
Training only

CE(logits, y)
AdamW / cosine

Argmax
Class ID

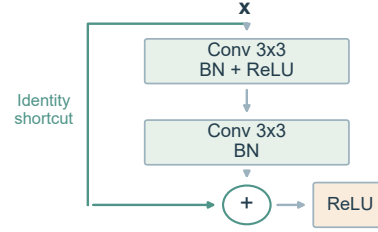
Val macro-F1 → selected checkpoint → held-out test

b ResNet18 feature hierarchy



C x H x W; stacked planes are schematic, not activation maps

c Residual block detail



Same-shape block shown; stage entries use
1x1 / stride-2 projection (Layers 2-4).